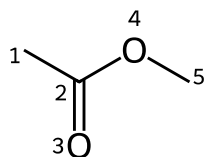
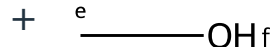
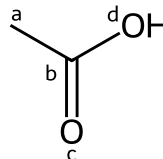
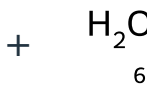


a One reaction, several atom correspondences

Methyl acetate + water → acetic acid + methanol



Reactant indices: 1-6



Product indices: a-f

b Growth and distinct placement branches

Local example: ethanol fragment in ethyl lactate.

seed

extend

saturate

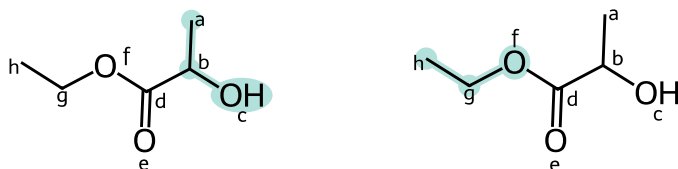
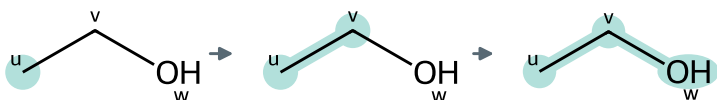


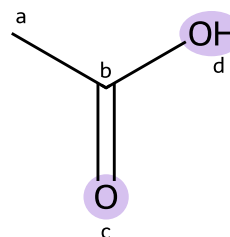
image {a,b,c}

image {f,g,h}

Different atom sets → different continuations (two shown).

c Compress matching-preserving shuffles

Single and double C-O bonds both pass at $\tau_{iso} = 1$.



R bond P bond

2 → 2 or 1

1 → 1 or 2

$|\Delta w| \leq 1$

$g = (c\ d)$ one coupled O exchange

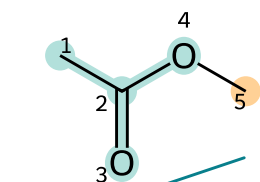
$\chi(1) = \chi(2)$: an intrinsic fragment symmetry.

Saved constraints decide which shuffles are allowed.

This is not formal-bond-order chemical symmetry.

d Branch again when fragments compete

Retain A, or let B claim the contacting oxygen 4.

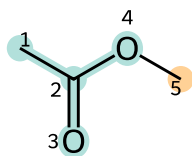


A = {1,2,3,4}

B starts at {5}

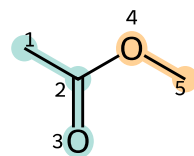
keep A

prioritize B



4 → d; 5 → e

cleave O4-C5

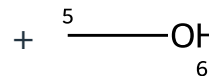
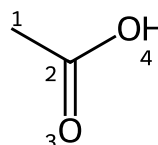


4 → f; 5 → e

cleave C2-O4

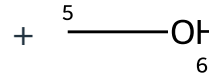
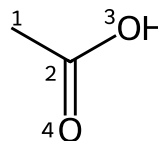
e Decode different bond-event patterns

Products labeled by their assigned reactant indices.



2 events

-(4,5), +(5,6)



4 events

Same changes, plus -(2,3), +(2,4)

O exchange changes events despite admissible matching.

f One unordered branch; retain its mapping families

The oxygen assignments in (c,e) share $\{(1,2,3,4) \leftrightarrow (a,b,c,d), 5 \leftrightarrow e, 6 \leftrightarrow f\}$, regardless of growth order.

Constructed heavy-atom illustration · formal bond weights · event threshold 0.5 · H transfers omitted